

you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please do *not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2026.tex` may be used as a “shell” for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 2, 3, and 4 below.

2 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by $\frac{1}{2}$ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow $\frac{1}{4}$ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors’ names are set in boldface, and each name is centered above the corresponding address. The lead author’s name is to be listed first (left-most), and the co-authors’ names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 4 regarding figures, tables, acknowledgments, and references.

3 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

3.1 Headings: second level

Second-level headings should be in 10-point type.

3.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

4 Citations, figures, tables, references

These instructions apply to everyone.

4.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

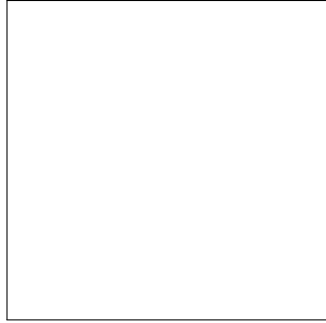


Figure 1: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

69 `http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

70 Of note is the command `\citet`, which produces citations appropriate for use in inline text. For
71 example,

72 `\citet{hasselmo}` investigated\dots

73 produces

74 Hasselmo, et al. (1995) investigated...

75 If you wish to load the `natbib` package with options, you may add the following before loading the
76 `neurips_2026` package:

77 `\PassOptionsToPackage{options}{natbib}`

78 If `natbib` clashes with another package you load, you can add the optional argument `nonatbib`
79 when loading the style file:

80 `\usepackage[nonatbib]{neurips_2026}`

81 As submission is double blind, refer to your own published work in the third person. That is, use “In
82 the previous work of Jones et al. [4],” not “In our previous work [4].” If you cite your other papers
83 that are not widely available (e.g., a journal paper under review), use anonymous author names in the
84 citation, e.g., an author of the form “A. Anonymous” and include a copy of the anonymized paper in
85 the supplementary material.

86 4.2 Footnotes

87 Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number¹
88 in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote
89 with a horizontal rule of 2 inches (12 picas).

90 Note that footnotes are properly typeset *after* punctuation marks.²

91 4.3 Figures

92 All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes.
93 The figure number and caption always appear after the figure. Place one line space before the figure
94 caption and one line space after the figure. The figure caption should be lower case (except for the
95 first word and proper nouns); figures are numbered consecutively.

96 You may use color figures. However, it is best for the figure captions and the paper body to be legible
97 if the paper is printed in either black/white or in color.

¹Sample of the first footnote.

²As in this example.

Table 1: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	≈ 100
Axon	Output terminal	≈ 10
Soma	Cell body	up to 10^6

98 4.4 Tables

99 All tables must be centered, neat, clean, and legible. The table number and title always appear before
100 the table. See Table 1.

101 Place one line space before the table title, one line space after the table title, and one line space after
102 the table. The table title must be lower case (except for the first word and proper nouns); tables are
103 numbered consecutively.

104 Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the
105 booktabs package, which allows for typesetting high-quality, professional tables:

106 <https://www.ctan.org/pkg/booktabs>

107 This package was used to typeset Table 1.

108 4.5 Math

109 Note that display math in bare TeX commands will not create correct line numbers for sub-
110 mission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You
111 really shouldn't be using \$\$ anyway; see [https://tex.stackexchange.com/questions/](https://tex.stackexchange.com/questions/503/why-is-preferable-to)
112 [503/why-is-preferable-to](https://tex.stackexchange.com/questions/503/why-is-preferable-to) and [https://tex.stackexchange.com/questions/40492/](https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath)
113 [what-are-the-differences-between-align-equation-and-displaymath](https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath) for more infor-
114 mation.)

115 4.6 Final instructions

116 Do not change any aspects of the formatting parameters in the style files. In particular, do not modify
117 the width or length of the rectangle the text should fit into, and do not change font sizes. Please note
118 that pages should be numbered.

119 5 Preparing PDF files

120 Please prepare submission files with paper size "US Letter," and not, for example, "A4."

121 Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or
122 Embedded TrueType fonts. Here are a few instructions to achieve this.

- 123 • You should directly generate PDF files using pdf_latex.
- 124 • You can check which fonts a PDF files uses. In Acrobat Reader, select the menu
125 Files>Document Properties>Fonts and select Show All Fonts. You can also use the pro-
126 gram pdffonts which comes with xpdf and is available out-of-the-box on most Linux
127 machines.
- 128 • xfig "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- 129 • The \bbold package almost always uses bitmap fonts. You should use the equivalent AMS
130 Fonts:

131 `\usepackage{amsfonts}`

132 followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also
 133 use the following workaround for reals, natural and complex:

```
134 \newcommand{\RR}{\mathbb{R}} %real numbers
135 \newcommand{\Nat}{\mathbb{N}} %natural numbers
136 \newcommand{\CC}{\mathbb{C}} %complex numbers
```

137 Note that `amssymb` is automatically loaded by the `amssymb` package.

138 If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

139 5.1 Margins in L^AT_EX

140 Most of the margin problems come from figures positioned by hand using `\special` or other com-
 141 mands. We suggest using the command `\includegraphics` from the `graphicx` package. Always
 142 specify the figure width as a multiple of the line width as in the example below:

```
143 \usepackage[pdftex]{graphicx} ...
144 \includegraphics[width=0.8\linewidth]{myfile.pdf}
```

145 See Section 4.4 in the `graphics` bundle documentation ([http://mirrors.ctan.org/macros/](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf)
 146 [latex/required/graphics/grfguide.pdf](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf))

147 A number of width problems arise when L^AT_EX cannot properly hyphenate a line. Please give LaTeX
 148 hyphenation hints using the `\-` command when necessary.

149 References

150 References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level
 151 heading for the references. Any choice of citation style is acceptable as long as you are consistent.
 152 It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the
 153 Reference section does not count towards the page limit.

154 [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In
 155 G. Tesauero, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp.
 156 609–616. Cambridge, MA: MIT Press.

157 [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the*
 158 *GENeral NEural Simulation System*. New York: TELOS/Springer-Verlag.

159 [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent
 160 synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-
 161 5262.

162 A Technical appendices and supplementary material

163 Technical appendices with additional results, figures, graphs, and proofs may be submitted with the
 164 paper submission before the full submission deadline (see above). You can upload a ZIP file for
 165 videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for
 166 the technical appendices.

167 Note: Think of the appendix as “optional reading” for reviewers. The paper must be able to stand
 168 alone without the appendix; for example, adding critical experiments that support the main claims
 169 to an appendix is inappropriate.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and introduction clearly state the three main contributions (state representation design, truncation-aware GAE, inference-time ensemble) and report the corresponding quantitative results (mean reward, programs beating -0z). All claims are directly supported by experiments in Section 4, including a cross-architecture evaluation (Section 4.5) where the proposed `pass_sequence` feature is integrated into a recurrent GRU baseline, demonstrating its architectural robustness and independently replicating the ablation finding across two systems developed in parallel.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 5 (Conclusion) includes a dedicated Limitations paragraph discussing the absence of GA/SA/MCTS baselines under controlled compute budget, the structural failure of the 30-pass action set on `linux-v0` kernel code, and the saturation of the current PPO design space.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[N/A\]](#)

Justification: This paper presents no theoretical results; all contributions are empirical.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: Section 4.1 specifies all PPO hyperparameters ($\epsilon=0.2$, $\gamma=0.99$, $\lambda=0.95$, `n_steps`, entropy coefficient, network architecture), the action set, reward space, episode length, and number of training episodes for both Phase 1 (MLP) and Phase 2 (GRU) architectures. The environment (CompilerGym `l1vm-v0`, `cbench-v1`) is publicly available under the Apache 2.0 license.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [\[No\]](#)

Justification: Code is not released with this submission due to double-blind anonymity requirements. The benchmark data (`cbench-v1`) is publicly available through CompilerGym ?. All hyperparameters, network architectures, and implementation details required for reproduction are fully specified in Section 4.1 and the PPO configuration table. Code will be released upon acceptance.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: Section 4.1 provides full training details for both architectures: Phase 1 uses a feed-forward 256×256 MLP over the full 124-pass action space (500 episodes per benchmark, NVIDIA RTX 5070 Ti), and Phase 2 uses a recurrent GRU over the curated 30-pass action space (300k steps, 8 parallel workers, Apple Mac mini M4 Pro). Hyperparameters were chosen via the systematic ablation studies in Sections 4.2–4.4, covering sequence window k , network capacity, n_steps , and entropy coefficient.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Error bars over multiple training seeds are omitted due to the high computational cost of the compiler environment. However, to establish statistical significance, Section 4.6 introduces a compute-budget-matched baseline (Random best-of-56) that gives the random policy $3.5 \times$ more rollouts than our PPO (56 vs. 16). Our learned policy outperforms this stronger baseline on all 23 benchmarks (avg 1.060 vs. 0.992), providing strong evidence that gains stem from policy knowledge rather than sampling variance. The ablation study further evaluates each configuration across all 21–23 cbench-v1 benchmarks, and the count of programs beating -Oz provides a distribution-free summary statistic.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Phase 1 ablation experiments (Sections 4.2–4.4) and the compute-budget comparison (Section 4.6) were conducted on a machine with an **NVIDIA RTX 5070 Ti** GPU (500 episodes per benchmark). Phase 2 ensemble training and cross-suite evaluation (Sections 4.7–4.8) were conducted on an **Apple Mac mini M4 Pro** (24 GB unified memory) using 8 parallel CPU workers, where each 300k-step training run completes in approximately 47 minutes. Cross-architecture validation (Section 4.5) uses the Phase 2 hardware for GRU configurations (v4, v7, v8).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: This work studies compiler optimization, a purely technical domain with no human subjects, sensitive data, or dual-use concerns. The research conforms with the NeurIPS Code of Ethics.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This work improves compiler optimization efficiency, which has broadly positive effects (reduced energy consumption, faster software). There are no foreseeable negative societal impacts from automated compiler pass ordering.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: This paper poses no misuse risks. The trained models are compiler optimization policies operating solely within the LLVM compiler infrastructure, with no dual-use potential.

270 **12. Licenses for existing assets**

271 Question: Are the creators or original owners of assets (e.g., code, data, models), used in

272 the paper, properly credited and are the license and terms of use explicitly mentioned and

273 properly respected?

274 Answer: [\[Yes\]](#)

275 Justification: CompilerGym ? and the cbench-v1 benchmark suite are cited in the paper

276 and used in accordance with the Apache 2.0 license. All other third-party dependencies

277 (PyTorch, Stable-Baselines3) are similarly open-source and used within their respective

278 license terms.

279 **13. New assets**

280 Question: Are new assets introduced in the paper well documented and is the documenta-

281 tion provided alongside the assets?

282 Answer: [\[N/A\]](#)

283 Justification: This paper does not release new datasets or pretrained model weights.

284 The primary contributions are algorithmic: the `pass_sequence` state representation, the

285 truncation-aware GAE implementation, and the inference-time ensemble strategy. Imple-

286 mentation details sufficient for re-implementation are provided in Sections 3 and 4.

287 **14. Crowdsourcing and research with human subjects**

288 Question: For crowdsourcing experiments and research with human subjects, does the pa-

289 per include the full text of instructions given to participants and screenshots, if applicable,

290 as well as details about compensation (if any)?

291 Answer: [\[N/A\]](#)

292 Justification: This paper does not involve crowdsourcing or human subjects.

293 **15. Institutional review board (IRB) approvals or equivalent for research with human**

294 **subjects**

295 Question: Does the paper describe potential risks incurred by study participants, whether

296 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

297 approvals (or an equivalent approval/review based on the requirements of your country or

298 institution) were obtained?

299 Answer: [\[N/A\]](#)

300 Justification: This paper does not involve human subjects research.

301 **16. Declaration of LLM usage**

302 Question: Does the paper describe the usage of LLMs if it is an important, original, or

303 non-standard component of the core methods in this research?

304 Answer: [\[N/A\]](#)

305 Justification: LLMs are not used in the core methodology of this work. The paper explic-

306 itly positions itself as a traditional DRL alternative to LLM-based compiler optimization

307 approaches (Section 2.2), and all policy networks are conventional MLP and GRU archi-

308 tectures trained with PPO.